

TAYLOR PENTZ

Senior Software Engineer – Python (FastAPI/Django), React, AWS

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Professional Summary

Senior Software Engineer with 10+ years of experience designing and delivering scalable enterprise applications across SaaS, fintech, and cloud-native environments using **Python, React, TypeScript, AWS, microservices, and distributed systems**. Strong background in building high-performance APIs, modernizing legacy platforms, leading cloud migrations, optimizing application performance, and delivering secure, production-grade software solutions across full software development lifecycles.

Technical Skills

- ❖ Programming Languages: Python, JavaScript, TypeScript, SQL, Bash
- ❖ Backend Technologies: Python 3.7–3.12, FastAPI, Django, Flask, Celery, SQLAlchemy, REST APIs, GraphQL, Microservices, Async Programming
- ❖ Frontend Technologies: React 16–19, TypeScript, Redux Toolkit, Next.js, HTML5, CSS3, Material UI, Tailwind CSS
- ❖ Cloud Platforms: AWS (EC2, Lambda, ECS, EKS, S3, RDS, CloudWatch, IAM, API Gateway), Azure
- ❖ Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB
- ❖ DevOps & CI/CD: Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Helm, ArgoCD
- ❖ Messaging & Event Streaming: Kafka, RabbitMQ, AWS SQS, SNS
- ❖ Testing: PyTest, Jest, React Testing Library, Cypress, Postman
- ❖ Monitoring & Observability: Datadog, Grafana, Prometheus, ELK Stack, CloudWatch
- ❖ Architecture: Microservices, Event-Driven Architecture, Distributed Systems, Domain Driven Design, Serverless, RESTful APIs
- ❖ Practices: Agile, Scrum, TDD, Code Reviews, CI/CD, Performance Optimization, System Design

Professional Experience

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 – Dec 2025

- Designed and implemented cloud-native backend services using **Python, FastAPI, PostgreSQL, and AWS**, supporting high-volume customer-facing applications that processed millions of API transactions monthly.
- Architected and developed scalable **microservices-based systems** using **Docker, Kubernetes, and AWS ECS**, improving deployment consistency and operational efficiency across distributed workloads.
- Built reusable frontend components using **React 18, TypeScript, Redux Toolkit, and Material UI**, reducing duplicated UI logic and improving developer productivity across teams.
- Modernized legacy monolithic services into containerized services, resolving API timeout issues and reducing average response times by **38%** through query optimization and asynchronous processing.

- Implemented background task processing using **Celery and Redis**, eliminating bottlenecks caused by synchronous workflows and improving processing throughput.
- Created secure authentication and authorization workflows using **JWT, OAuth2, and AWS IAM**, strengthening application security for enterprise customers.
- Led migration efforts from older deployment pipelines to **GitHub Actions and Terraform**, enabling infrastructure standardization and faster releases.
- Resolved production issues involving database connection exhaustion, API throttling, and container scaling problems using proactive monitoring and tuning.
- Implemented observability using **CloudWatch, Datadog, and Grafana**, improving incident detection and reducing troubleshooting effort.
- Collaborated with product managers and engineering leadership on architecture reviews, roadmap planning, and technical mentoring for junior engineers.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Developed enterprise applications using **Python, Django, Flask, React, and PostgreSQL** for financial and data-driven platforms.
- Designed and maintained RESTful APIs supporting multiple frontend applications and third-party integrations.
- Built responsive frontend applications using **React 16–18, Redux, and TypeScript**, improving application usability and maintainability.
- Migrated legacy frontend modules into reusable React component architecture, reducing maintenance complexity.
- Optimized slow database operations through indexing strategies, caching, and query tuning, improving backend processing performance by **32%**.
- Developed asynchronous workflows using **Kafka, Celery, and Redis**, enabling scalable event processing.
- Implemented automated CI/CD pipelines using **Jenkins, Docker, and Kubernetes**, reducing deployment overhead.
- Addressed recurring production issues involving memory leaks, stale cache problems, and API retries by introducing improved monitoring practices.
- Worked closely with QA teams implementing automated testing using **PyTest and Jest** to improve release quality.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Developed backend services using **Python, Flask, SQLAlchemy, and MySQL** supporting customer-facing business applications.
- Built frontend interfaces using early **React versions and JavaScript**, helping transition older UI implementations.
- Created reusable API layers for mobile and web applications using REST principles.
- Resolved performance issues related to large dataset processing through pagination and query optimization.
- Implemented database migration scripts and schema updates supporting product growth.
- Participated in migration efforts from tightly coupled architectures toward service-based systems.

- Improved release reliability by introducing automated testing workflows and code review standards, contributing to approximately **25%** fewer post-release issues.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Supported internal application development using **Python, JavaScript, SQL, and web technologies**.
- Assisted with debugging production defects and improving application stability.
- Created scripts for automation tasks and developer productivity improvements.
- Participated in testing, code reviews, and software delivery processes.
- Collaborated with senior engineers on feature implementation and documentation.
- Created internal dashboards supporting engineering visibility and reporting.
- Improved code maintainability through refactoring and cleanup initiatives.

Projects

Cloud-Native Customer Analytics Platform

- Developed a distributed analytics platform using **FastAPI, React, Kafka, PostgreSQL, Redis, and AWS**, enabling near real-time customer insights and scalable event processing across multiple business domains.

Enterprise Workflow Automation System

- Built a workflow orchestration platform using **Python, React, Celery, Docker, Kubernetes, and Terraform**, automating business operations and significantly reducing manual operational effort.

Certifications

- **AWS Certified Developer – Associate**
- **AWS Certified Solutions Architect – Associate**
- **Microsoft Azure Fundamentals**
- **Certified Kubernetes Application Developer (CKAD)**
- **Professional Scrum Master I (PSM I)**

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)